



COMPUTER

- Assume 'x' represents negation of x the Boolean function $x'y' + xy + x'y$ is equivalent to?
(a) $x' + y$ (b) $x + y$ (c) $x + y'$ (d) $x' + y'$
- The memory unit which directly communicates with the CPU is known as
(a) Primary Memory (b) Secondary Memory
(c) Shared Memory (d) Auxiliary Memory
- Dynamic RAM consumespower andthan Static RAM
(a) More, Faster (b) More, slower
(c) Less, Slower (d) Less, Faster
- The binary equivalent of $(234.125)_{10}$
(a) $(11101010.101)_2$ (b) $(10101010.011)_2$
(c) $(11101010.001)_2$ (d) $(10101110.011)_2$
- Determine the octal equivalent of $(432267)_{10}$?
(a) $(432267)_8$ (b) $(346731)_8$
(c) $(2164432)_8$ (d) None of these
- One Exabyte is equal to...
(a) 1018 Bytes
(b) 1 Zetta Bytes divided (/) by one thousand
(c) 1 Peta Bytes multiplied (x) by one thousand
(d) All of the above
- Consider the following circuit.

How many minimum numbers of two input NAND gates are required to design the above circuit?
(a) 6 (b) 4 (c) 5 (d) 3
- The time required for fetching and execution of one simple machine instruction is known as
(a) Delay time (b) CPU cycle
(c) Real time (d) Seek Time
- The equivalence of given expression $x + x'y$ with Boolean theorem is.....
(a) x (b) $x + y$ (c) x' (d) 0
- The logic XOR operation of $(4AC0)_{16}$ and $(B53F)_{16}$ results
(a) AACB (b) 0000 (c) FFFF (d) ABCD

ANSWER

1.	A	2.	A	3.	C	4.	C	5.	D
6.	D	7.	B	8.	B	9.	B	10.	C

SOLUTION

1. (a)

$$\begin{aligned}
 &x'y' + xy + x'y \\
 &= xy + x'y' + x'y \\
 &= xy + x'(y' + y) \\
 &= xy + x'1 \quad [\because A + A' = 1] \\
 &= xy + x' \\
 &= (x+x')(y + x') \quad [\because (A + BC) = (A+B)(A+C)] \\
 &= 1.(y + x') \quad [\because A + A' = 1] \\
 &= x'y
 \end{aligned}$$

2. (a)

The memory unit which directly communicates with the CPU is known as primary memory. Primary memory or main memory refers to physical memory that is internal to the computer. Every program you execute and every file you access must be copied from a storage device into main memory.

3. (c)

Dynamic RAM consumes less power and slower than static RAM, because of the needed of charging and discharging of capacitor in DRAM. It consumes less power than SRAM also slow too.

4. (c)

First convert $(234)_{10}$ to binary

2	234	0
2	117	1
2	58	0
2	29	1
2	14	0
2	7	1
2	3	1
2	1	1
	0	

Here, $(234)_{10} = (11101010)_2$

Now, convert $(0.125)_{10}$ to Binary

0.125×2	0.250	0
0.250×2	0.500	0
0.500×2	1.000	1

Here, $(0.125)_{10} = (.001)_2$

Hence, $(234.125)_{10} = (11101010.001)_2$

5. (d)

8	432267	3
8	54033	1
8	6754	2
8	844	4
8	105	1
8	13	5
8	1	1
	0	

Hence, $(432267)_{10} = (1514213)_8$

6. (d)

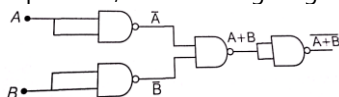
Exabyte is a unit to measure memory which is equal to 10^{18} Bytes, $1000 \times 1 \text{ PB}$ and $\frac{128}{1000}$.





7. (b)

We need to create NOR gate using NAND gates using the boolean expression, the following diagram is possible.



So, minimum number of two input NAND gates are 4.

8. (b)

The time required for the fetching and execution of one simple machine instruction is CPU cycle.

The speed of a computer processor or CPU, is determined by the clock cycle, which is the amount of time between two pulses of an oscillator. The higher number of pulses per second, the faster the computer processor will be able to process information.

9. (b)

$$\begin{aligned} x + x'y &= x.1 + x'y \\ &= x(1+y) + x'y \quad [\because A.1 = A] \\ &= x + xy + x'y \quad [\because A' + A = 1] \\ &= x + y(x + x') \\ &= x + y.1 \\ &= x + y \end{aligned}$$

10. (c)

Binary equivalent of

$$(4ACO)_{16} = 0100101011000000$$

Binary equivalent of

$$(B53F)_{16} = 1011010100111111$$

Now, perform the XOR operation on $(4ACO)_{16}$ and $(B53F)_{16}$
 $(1111111111111111)_2$

It is equivalent of $(FFFF)_{16}$.

